

Michael J Lalonde

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Senior Engineering Leader with extensive experience scaling high-volume, low-latency systems. Proven track record of nurturing both engineering talent and software platforms from conception through production. Combines hands-on technical expertise with team leadership to drive innovation, performance, and engineering excellence.

Experience

MachineQ, a Comcast Company

Philadelphia, PA

Director, Product Engineering

2018–Present

- Designed and implemented comprehensive platform optimizations that cut cloud costs by 80%, reduced data processing time from seconds to milliseconds, and enabled throughput scaling beyond 100x of original capacity.
- Established comprehensive code quality standards and peer review processes that reduced production incidents from monthly to nearly eliminated (>95% reduction), while cultivating a culture of collaborative problem-solving.
- Developed backward compatibility protocols enabling confident weekly deployments that accelerated feature delivery with zero customer disruptions.
- Transformed engineering culture through strategic focus on ownership and technical excellence, resulting in promotions for 50% of team members and conversion of contractors to full-time roles.

Comcast

Philadelphia, PA

Engineering Manager

2014–2018

- Directed team of Java engineers building mission-critical systems with 99.99% uptime for infrastructure serving 25+ million Comcast customers.
- Implemented autonomous team structure and decision-making frameworks that accelerated feature delivery velocity by 25% while improving code quality metrics.
- Championed DevOps transformation that boosted production stability by 60% and slashed deployment times by 50%, enabling near daily releases.
- Led strategic adoption of Golang and additional modern language stacks that resolved persistent scaling bottlenecks and reduced infrastructure costs by 40%.

Comcast

Philadelphia, PA

Senior Software Engineer

2008–2014

- Designed the architecture and delivered multiple high-performance platforms using Java, Golang, Erlang, and Clojure.
- Optimized service architecture to handle 10x traffic growth while reducing hardware footprint by 25%.
- Created comprehensive on-boarding guides and mentored junior engineers, improving team productivity.
- Led adoption of continuous integration practices and automated testing that reduced regression bugs by 70%.
- Developed systematic troubleshooting methodologies that reduced mean time to resolution by 65%.

Technical Skills

Languages: Go, SQL, Python, JavaScript, C, HTML/CSS

Infrastructure: Docker, Kubernetes, Terraform, Azure Cloud, Hashicorp Vault, Nginx, Apache

Data Systems: PostgreSQL, Redis, Kafka, SQLite

Tools & Observability: Git, Grafana, KQL, Make, Emacs, Node

Previous Experience: Java, Scala, Erlang, Lisp, PHP, .NET, Oracle

Projects

Animated Brewing

Coatesville, PA

Automated Keg Washing System

2020

Designed and built a cost-effective automated keg washing system for a local craft brewery, reducing manual labor by 80% and cutting cleaning time in half compared to traditional methods.

- Programmed a Teensy microcontroller in C to control automated valve sequences and cleaning cycles.
- Engineered complete system architecture including piping diagrams, electrical schematics, and component selection.
- Implemented intuitive touchscreen interface with real-time status monitoring and multiple washing programs.
- Delivered solution at 15% of commercial system cost while maintaining industry cleaning standards.

Ethereum Network

Home

Ethereum Validator Node

2019

Built and deployed an Ethereum validator node for the Beacon Chain, achieving 99.8% uptime and consistent attestation performance.

- Configured optimized hardware and networking setup for 24/7 reliable operation.
- Developed custom monitoring solution using Telegram bot for real-time alerts and performance tracking.
- Gained deep technical understanding of blockchain consensus protocols, peer discovery, and distributed systems.

Additional Experience

SunGARD

Philadelphia, PA

Software Engineer

2003–2008

- Designed and implemented network monitoring systems processing millions of events daily.
- Managed a data warehouse solution for long-term metrics storage with automated rollups and aggregations.
- Created custom correlation engine that reduced false-positive alerts by 85%.
- Developed customer-facing dashboards providing real-time visibility into network performance.

Cadient Group

Conshohocken, PA

IT Engineer

2002–2003

- Led infrastructure planning and implementation for growing startup, coming in 10% under budget.
- Designed and deployed complete network environment including domain, email, web, and backup services.
- Successfully migrated all users and data to new systems with minimal downtime.

Qwest Global Services

Philadelphia, PA

Solutions Engineer

2000–2002

- Served as technical lead on critical client projects, delivering on-time and to specification.
- Developed custom applications for internal use and client deployment using multiple technologies.

IKON Office Solutions

Malvern, PA

Web Developer

1999–2000

- Led design and implementation of corporate Intranet serving 40,000+ employees globally.
- Created custom web applications and content management solutions for company divisions.